

K-5 Maintenance

This section describes maintenance procedures and guidelines for the FastBack® Model 260E-G2. The following topics are discussed in this section.

- Recommended Maintenance Intervals
- Cleaning
- End-gate and Mid-gate Cleaning and Maintenance
- Removing and Installing Enclosure Side Panel
- Motor Belt Replacement and Tensioning
- Motor Replacement
- Transmission Replacement
- Eccentric Belt Replacement and Tensioning
- Checking / Reversing the Direction of Product Flow
- Adding / Removing Counterweight Plates
- VFD Replacement

Recommended Maintenance Intervals

Table 5-1 lists recommended maintenance intervals for the FastBack® Model 260E-G2 and the type of maintenance to be performed at each interval.

Interval	Recommended Maintenance
Following each cleaning	Check Photoeye calibration
Following each shift or pan replacement	Check Photoeye calibration
Weekly	Check tightness of enclosure latches Monitor for any unusual noises coming from the drives Check tightness of locknuts on threaded stems of articulating feet
Every 6 months	Check all belts Check tightness of pan connection bolts
Every 12 months	Check all belts and re-tension as required

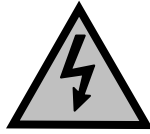
Table 5-1 – Recommended Maintenance Intervals

K-5 – MAINTENANCE

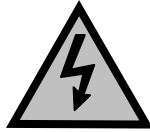
FastBack® Model 260E-G2 Technical Manual

Cleaning

Observe the following guidelines when cleaning the FastBack® Model 260E-G2.



Multiple voltage sources may be present, including control wiring from the scales, PLCs, or other devices. Use the same safety precautions as with any equipment powered by electricity.



Turn off all power at the source before applying any water.



Limit high-pressure water to product handling areas of the conveyor. Never direct high-pressure water at the Photoeyes, drive assembly enclosure, or utility enclosure.



Do not use abrasives, scouring powders, or any product that may scratch the equipment surface.

End-gate and Mid-gate Cleaning and Maintenance

This procedure describes how to clean the End-gates and Mid-gates of your FastBack® 260E-G2.

Tools Needed:

- small spatula
- cleaning supplies

Step	Description
------	-------------

- | | |
|---|--|
| 1 | Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position. |
|---|--|

Turn off the pressurized air supply to the FastBack®.



WARNING

To prevent the FastBack® from accidentally starting during this procedure, lockout / tag out before servicing.

- | | |
|---|--|
| 2 | Disconnect the air fittings found on the air fitting bracket of the drive arm (see Photo 5-1). |
|---|--|

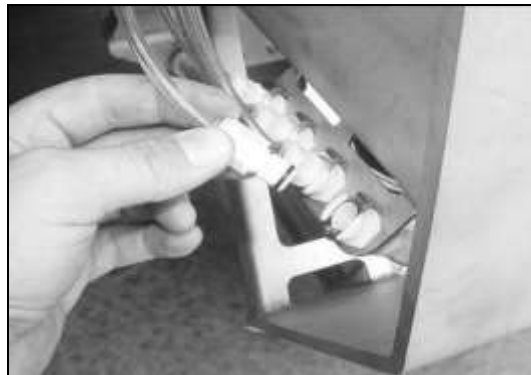


Photo 5-1 – Disconnect the air fittings

- 3 Clean the End-gate and pan. With the air lines disconnected, the end-gate tray should easily slide forward/backward to allow access to all surfaces.
Using the small spatula, clean all caked-on areas of product found on the End-gate and pan surfaces.
- 4 Clean the Mid-gate using the small spatula. Clean all caked-on areas of product found around the edges of the Mid-gate as shown in Photo 5-2.



Photo 5-2 – Cleaning the edges of the Mid-gate

Using the small spatula, loosen all caked-on areas of product found within the UHMW guide strips of the Mid-gate. Once the product is loose, it can easily be removed using your finger or a cleaning rag (see Photo 5-3).



Photo 5-3 – Removal of caked product from UHMW guide strips

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

Using the small spatula, clean all caked-on areas of product found where the Mid-gate surface meets the pan surface as shown in Photo 5-4.



Photo 5-4 – Cleaning where the Mid-gate surface meets the pan

- 5 Connect the air fittings located on the drive arm bracket in sequence to the gate fittings by snapping the quick disconnect in until it clicks (see Photo 5-5)



Photo 5-5 – Click air fitting into place

- 6 Restore power and pressurized air to the FastBack®.
- 7 Check gate motion. Ensure that the gates function in the proper sequence. If the gates are functioning in reverse, turn off the pressurized air supply to the FastBack® and switch the air lines as described in step number 5 of this procedure.

Note: *Switch UHMW guide strips found on Mid-gates from one side of the gate to the other. Sometimes, wear will be more pronounced on one end of the guide strips. Therefore, switching them around can extend the useful life by exposing the gate's contact surface to a new track surface.*

Removing and Installing Enclosure Side Panel

This procedure describes how to remove the drive assembly enclosure side panel.

Tools Needed:

- 3/8-inch [10 mm] standard screwdriver

Step	Description
------	-------------

- | | |
|---|--|
| 1 | Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position. |
|---|--|



WARNING

To prevent the FastBack® from accidentally starting during this procedure, lockout / tag out before servicing.

Moving parts are located behind the enclosure side panel. Do not operate with cover or enclosure side panel removed. Exercise caution if turning the mechanisms by hand to avoid crushing or pinching injuries.

- | | |
|---|---|
| 2 | Using a 3/8-inch [10 mm] standard screwdriver, turn the three (3) quarter-turn latches clockwise or counterclockwise (see Photo 5-7 for turning direction) to release the enclosure side panel from the enclosure |
|---|---|



Photo 5-6 – Turn clockwise or counterclockwise to release latches. Latch shown is in the closed position.



Photo 5-7 – Turning directions to remove panel

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 3 Pull the enclosure side panel away from the enclosure (see Photo 5-8).



Photo 5-8 – Enclosure side panel removed from drive unit

- 4 Inspect all three (3) hook latches to make sure that they are in good condition and no parts are loose (see Photo 5-9).



Photo 5-9 – Enclosure latch

- 5 Install the enclosure side panel by inserting the upper lip of the panel into the enclosure opening and then pressing the bottom of the panel against the chassis.

While pushing against the enclosure side panel, turn the three (3) quarter-turn latches clockwise or counterclockwise (see Photo 5-10 for turning direction) using a 3/8-inch [10 mm] standard screwdriver to seal the enclosure side panel to enclosure frame.

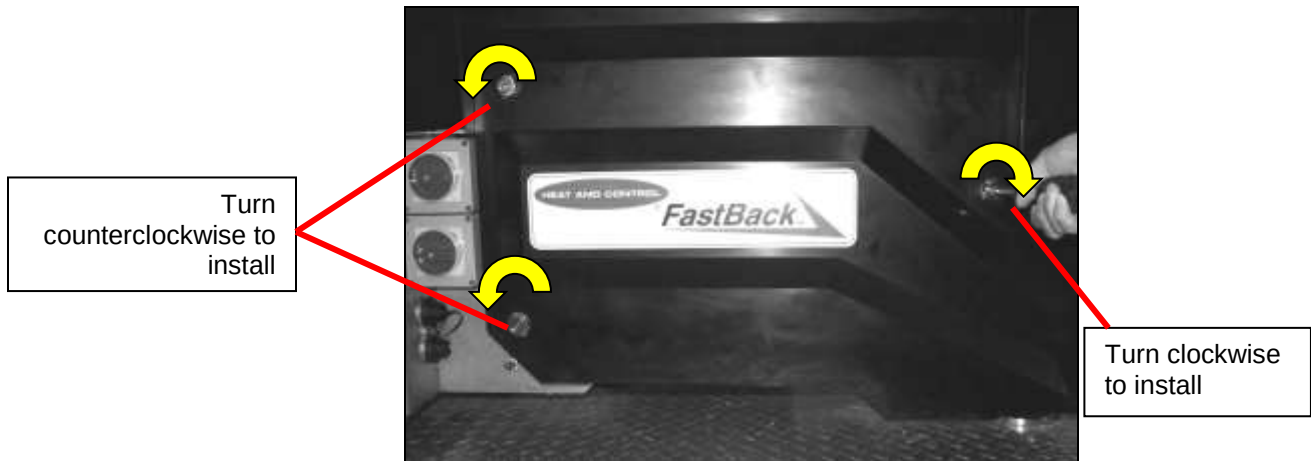


Photo 5-10 – Turning directions to install panel

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

Motor Belt Replacement and Tensioning

This procedure describes how to replace and properly tension the Motor Belt.

Tools Needed:

- 3/8-inch [10 mm] standard screwdriver
- socket wrench
- torque wrench set at 20 ft-lbs [27 N-m]
- 19 mm [3/4-inch] socket

Step	Description
------	-------------

- | | |
|---|--|
| 1 | Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position. |
|---|--|



WARNING

To prevent the FastBack® from accidentally starting during this procedure, lockout / tag out before servicing.

Moving parts are located behind the enclosure side panel. Do not operate with cover or enclosure side panel removed. Exercise caution if turning the mechanisms by hand to avoid crushing or pinching injuries.

- | | |
|---|--|
| 2 | Remove the enclosure side panel as described in “Removing and Installing Enclosure Side Panel”. |
| 3 | Loosen the motor plate nuts, using a 19 mm [3/4-inch] socket wrench, loosen four (4) motor plate nuts one-half turn from tight (see Photo 5-11). |

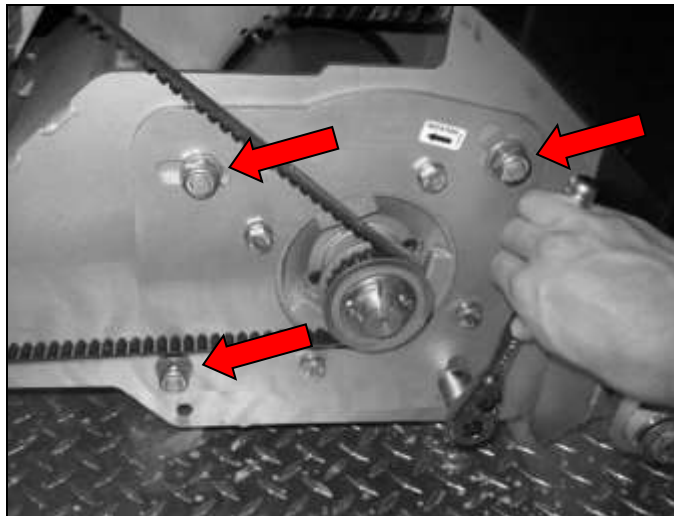


Photo 5-11 – Loosen motor plate nuts

Note: Skip steps 4 through 6 if re-tensioning the belt only.

- 4 Move the motor counterclockwise toward the transmission via the motor plate. Pinching the motor belt together will shift the motor in the correct direction (see Photo 5-12).

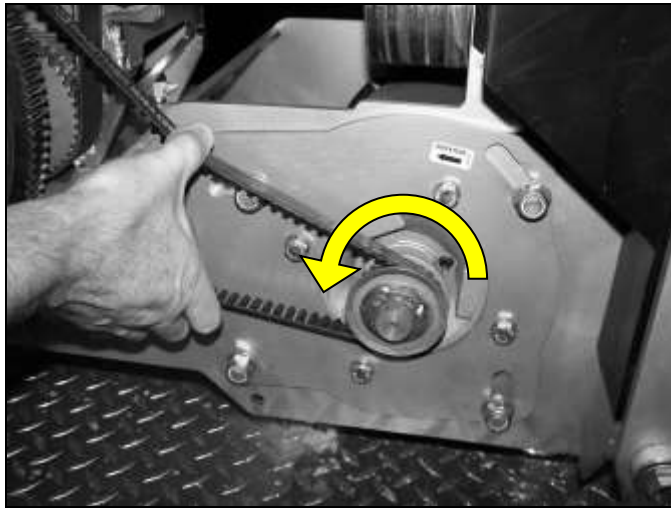


Photo 5-12 – Pinching the motor belt together

While holding the motor plate up, lock it down in place using one (1) of the four (4) motor plate nuts (see Photo 5-13).



Photo 5-13 – Motor plate secured in UP position

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 5 Remove and inspect the motor belt. Remove the motor belt by first removing it from the larger sprocket (see Photo 5-14). Once removed, inspect the belt for cracks or excessive wear. Replace if needed. See Appendix B for the belt's part number.

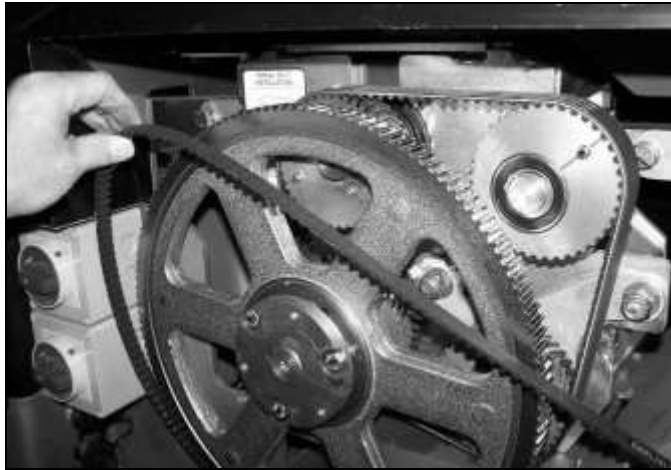


Photo 5-14 – Remove motor belt from larger sprocket first

- 6 Replace the belt by first placing it around motor sprocket and then the main drive sprocket.
Loosen the nut holding the motor plate one-half turn and turn the main drive sprocket by hand three full rotations in a counterclockwise direction to ensure proper tracking (see Photo 5-15).

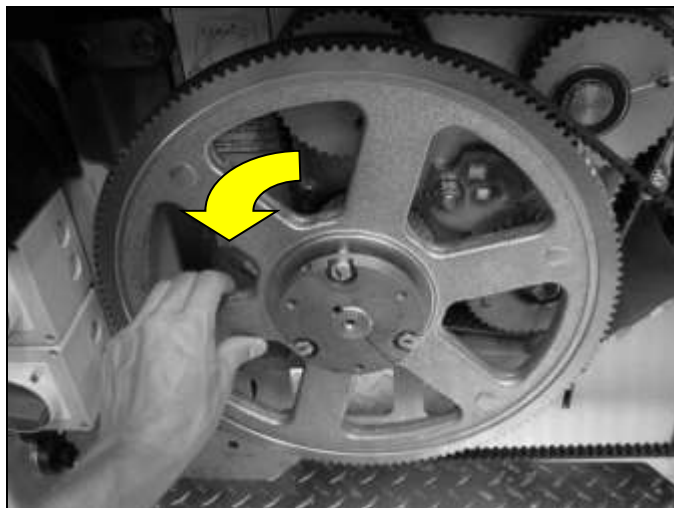


Photo 5-15 – Manually rotating the mechanism counterclockwise

The metal sprocket teeth should show on both sides of the belt as in Photo 5-16.



Photo 5-16 – Sprocket tooth margin both sides of belt

- 7 Tension the motor belt by pushing down on the motor sprocket. Ensure that the motor plate is not bound up and that the belt tension is taut. If additional tension is needed on the belt, use a socket wrench on the motor plate nuts to apply clock-wise torque to the system (see Photo 5-17). Do not over-tension the motor belt, as this will cause premature wear.

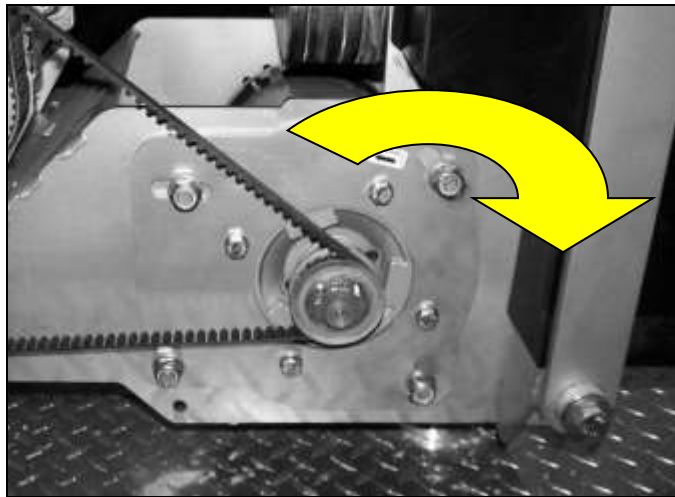


Photo 5-17 – Pivot motor away from transmission to tension belt

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 8 Using a 19 mm [3/4-inch] socket and a torque wrench tighten the four (4) motor plate nuts to 20 ft-lbs [27 N-m] (see Photo 5-18).



Photo 5-18 – Tighten motor plate with torque wrench

- 9 Install the enclosure side panel as described in “Removing and Installing Enclosure Side Panel”.

Motor Replacement

This procedure describes how to replace the motor and sprocket assembly.

Tools Needed:

- 3/8-inch [10 mm] standard screwdriver
- socket wrench
- 19 mm [3/4-inch] socket
- 14mm [9/16-inch] socket
- small machinist's square
- straight edge
- torque wrench set at 55 in-lbs [6.2 N-m], 20 ft-lbs [27 N-m]
- 1/4-inch [6.35 mm] socket
- 1/4-inch [6.35 mm] box wrench
- 1/8-inch [3.18 mm] Allen wrench

Step	Description
------	-------------

- | | |
|---|--|
| 1 | Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position. |
|---|--|



WARNING

To prevent the FastBack® from accidentally starting during this procedure, lockout / tag out before servicing.

Moving parts are located behind the enclosure side panel. Do not operate with cover or enclosure side panel removed. Exercise caution if turning the mechanisms by hand to avoid crushing or pinching injuries.

- | | |
|---|---|
| 2 | Remove the enclosure side panel as described in "Removing and Installing Enclosure Side Panel". |
| 3 | From underneath the chassis, locate the motor junction box. Using a 1/4-inch [7 mm] socket, open the motor junction box (see Photo 5-19). |



Photo 5-19 – Open motor junction box

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

Disconnect the motor leads from the motor junction box (see Photo 5-20).

Note: *Mark the black, red, and white inverter lead pairings before disconnecting.*



Photo 5-20 – Disconnect motor leads

Pull the cord grip out of motor junction box with leads still attached.

- 4 Remove the motor belt as described in “Motor Belt Replacement and Tensioning”.
- 5 Remove motor plate nuts, using a 19 mm [3/4-inch] socket wrench. Remove the four (4) motor plate nuts that attach the motor plate to the chassis (see Photo 5-21).

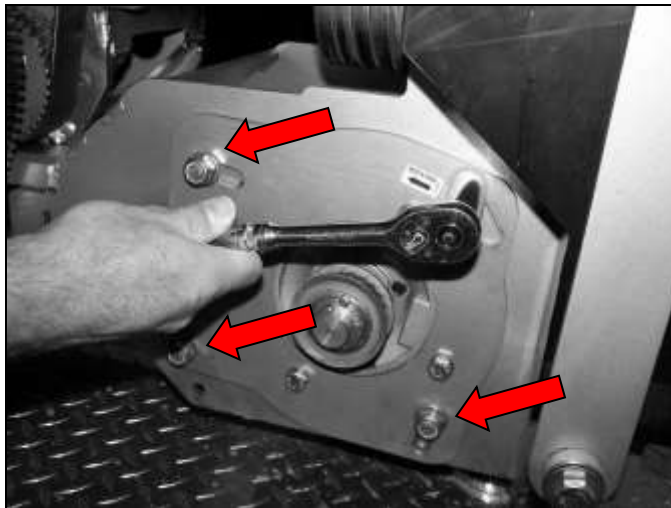


Photo 5-21 – Remove motor plate nuts

- 6 Remove the motor assembly by sliding the motor out through the compartment opening (see Photo 5-22). It may be necessary to twist the motor to clear the junction box protrusion.



Photo 5-22 – Remove the motor from the enclosure

- 7 Remove the motor plate from the motor. Using a 14 mm [9/16-inch] socket wrench, remove the four (4) bolts attaching the motor to the motor plate (see Photo 5-23).

Note: *The motor is mounted to the motor plate with four (4) 3/8-16 x 1-inch bolts.*

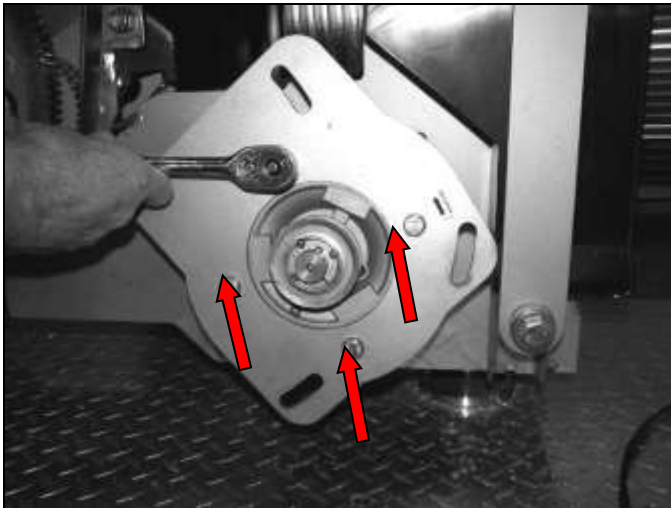


Photo 5-23 – Remove the motor mounting

K-5 – MAINTENANCE

- 8 Remove the motor drive sprocket. Place the straight edge against the motor drive sprocket. Use the machinist's square to measure the distance of this surface to the front surface of the motor (see Photo 5-24). Noting this measurement allows for easier alignment during reassembly.

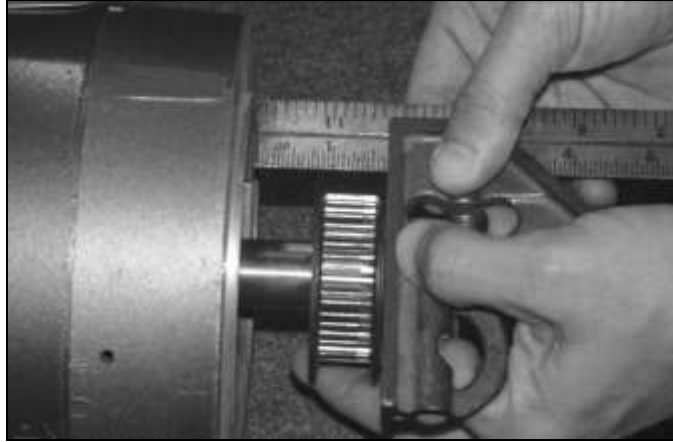


Photo 5-24 – Measuring sprocket distance to front of the motor

Using a 1/8-inch [3.18 mm] allen wrench, remove the set screws from the sprocket.

Using a 1/8-inch [3.18 mm] allen wrench, replace one (1) of the set screws in the unused hole and turn it carefully to push the sprocket off the bushing. The sprocket and bushing should be free to slide off the motor shaft.

- 9 Place drive sprocket on the new motor. Using the measurement from step number 8, and making sure the key is installed on the motor drive shaft, reinstall the motor drive sprocket.
Using the 1/8-inch [3.18 mm] allen wrench and a torque wrench set at 55 in-lbs [6.2 N-m], install the motor drive sprocket on the new motor drive shaft (see Photo 5-25).



Photo 5-25 – Attaching drive sprocket to new motor

- 10 Attach the motor plate to the motor, using a 14 mm [9/16-inch] socket wrench, reinstall the four (4) bolts attaching the motor plate to the new motor (see Photo 5-26).

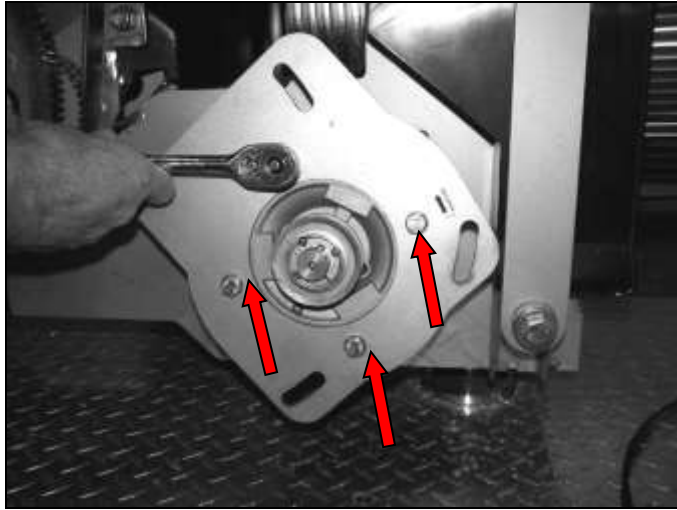


Photo 5-26 – Reinstall the motor plate on new motor

- 11 Slide the new motor into the chassis opening, ensuring that the motor junction box is in the 7 o'clock position, and realign the motor plate (see Photo 5-27). Reinstall the four (4) bolts and lock washers attaching the motor plate to the chassis.

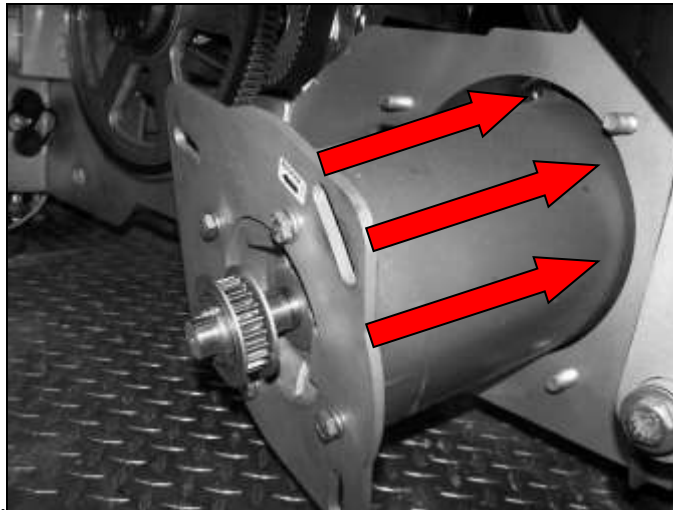


Photo 5-27 – Insert motor with junction box

- 12 Reinstall the motor belt.
Reinstall the motor belt as described in "Motor Belt Replacement and Tensioning" beginning with step 6.

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 13 Rewire the motor, from underneath the chassis, reinstall the cord grip in the new motor junction box and rewire the motor leads as previously configured.

Refer to motor wiring diagram found on motor housing or junction box lid if necessary (see Photo 5-28).



Photo 5-28 – Wiring diagram found on motor

Replace the motor junction box lid and tighten the four (4) attaching screws using a 1/4-inch [7 mm] box wrench.

- 14 Verify motor direction.
Restore power to the FastBack® and ensure that the motor turns in a counterclockwise direction as indicated by the “ROTATION” sticker on the face of the motor plate.



Do not perform this step with the motor belt in place.

Note: After verifying the motor direction, disconnect the power to the FastBack® and lockout / tag out before continuing.

- 15 Install the enclosure side panel as described in “Removing and Installing Enclosure Side Panel”.

Transmission Replacement

This procedure describes how to remove the transmission for maintenance or replacement of the transmission assembly.

Tools Needed:

- 3/8-inch [10 mm] standard screwdriver
- #2 Phillips head screwdriver
- socket wrench
- 14 mm [9/16-inch] socket
- small machinist's square
- torque wrench set at 15 ft-lbs [20 N-m], 20 ft-lbs [27 N-m] and 50 ft-lbs [68 N-m]
- 10-inch [254 mm] long 1/2-inch [12.7 mm] drive socket extension
- socket universal joint
- 19 mm [3/4-inch] deep well socket
- 19 mm [3/4-inch] open-ended wrench
- 1/2-inch [13 mm] socket

Step	Description
1	Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position.



WARNING

To prevent the FastBack® from accidentally starting during this procedure, lockout / tag out before servicing.

Moving parts are located behind the enclosure side panel. Do not operate with cover or enclosure side panel removed. Exercise caution if turning the mechanisms by hand to avoid crushing or pinching injuries.

- | | |
|---|---|
| 2 | Remove the enclosure side panel as described in "Removing and Installing Enclosure Side Panel". |
| 3 | Remove the motor belt as described in "Motor Belt Replacement and Tensioning". |

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 4 Remove the drive sprocket bushing.
Measure the distance from the drive sprocket bushing flange to the front face of the transmission using a machinist's square (see Photo 5-29). This measurement will be used during reassembly.



Photo 5-29 – Measure distance from bushing to transmission

Using a 1/2-inch [13 mm] socket wrench, remove the three (3) bolts and lock washers found on the drive sprocket bushing.

Using the same wrench, install the three bolts into the tapped holes in the bushing flange (see Photo 5-30). By incrementally turning each bolt in sequence around the bushing, the bushing will press itself away from the drive sprocket and disengage from the shaft.



Photo 5-30 – Remove drive sprocket bushing

- 5 Remove the drive sprocket the drive shaft.

Note: *The sprocket and bushing weigh approximately 22 lbs [10 kg].*

Once the bushing is free of the drive shaft, tilt the top towards you and lift out as shown in Photo 5-31.

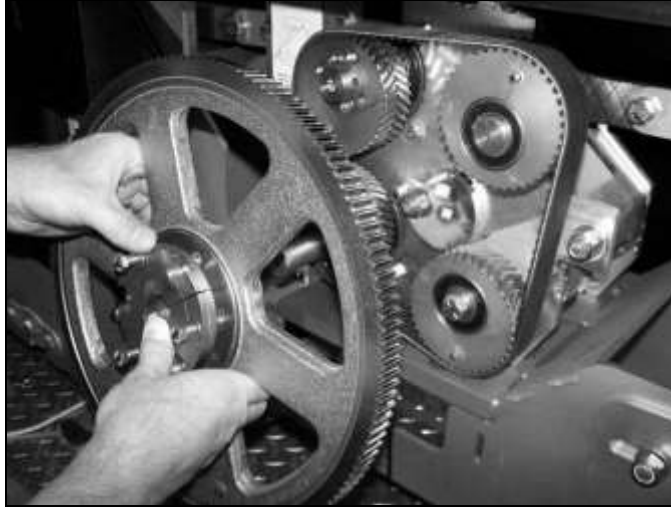


Photo 5-31 – Lift out drive sprocket

- 6 Remove the bolts from the pan and counterweight connecting rods located to the left and right of the transmission using a 19 mm [3/4-inch] socket wrench with extension (see Photo 5-32).

Note: *Rotate the eccentric sprocket system by hand to achieve the greatest access to the bolts.*

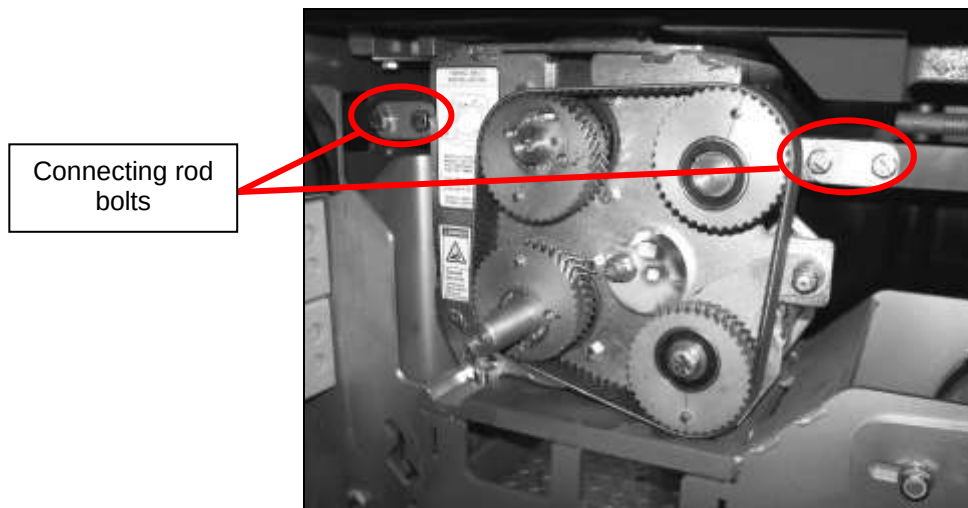


Photo 5-32 – Connecting rod bolts

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

For the far-left bolt, a socket universal joint must be used (see Photo 5-33).



Photo 5-33 – Using socket universal joint to remove far-left bolt

Lower the transmission side connecting rod halves by hand so that they rest on the transmission frame.

- 7 Remove the bottom attachment bolts. Using a 19 mm [3/4-inch] wrench, remove the four (4) bolts from the underside of the transmission (see Photo 5-34).

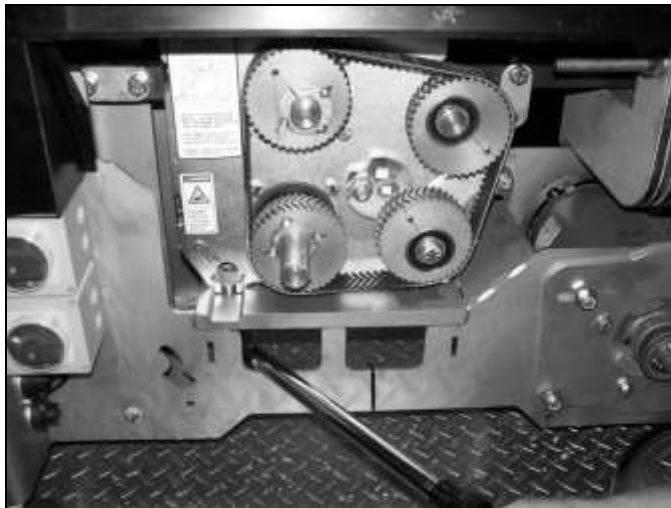


Photo 5-34 – Remove bolts from underside of transmission

- 8 Install the Transmission Removal Plate. Insert protrusions of transmission removal plate into the vertical slots of the chassis plate (see Photo 5-35).



Photo 5-35 – Install transmission removal plate

Push down securely until the plate is level (see Photo 5-36).



Photo 5-36 – Transmission removal plate shown in level position

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 9 Slide the transmission out onto the transmission removal plate (see Photo 5-37).

Note: *The transmission weighs approximately 114 lbs [52 kg].*



Photo 5-37 – Place transmission on transmission removal plate

Note: *If only re-tensioning or replacing the eccentric timing belt, proceed to “Eccentric Belt Replacement and Tensioning” upon completion of this step.*

- 10 With the transmission removal plate installed, slide the new transmission into place until it rests against the pin on the mounting plate (see Photo 5-38). Make sure the transmission is properly oriented and that the two bottom transmission tabs are seated in the groove on the left side of the plate.



Photo 5-38 – Install transmission

- 11 Remove the transmission removal plate. Tilt the transmission removal plate upward to dislodge the tabs from the chassis slots and pull outward to remove (see Photo 5-39).



Photo 5-39 – Remove transmission removal plate

- 12 Reinstall the bottom attachment bolts. Using a 19 mm [3/4-inch] socket and a torque wrench set to 50 ft-lbs [68 N-m], reinstall the four (4) bolts found on the underside of the transmission (see Photo 5-40).

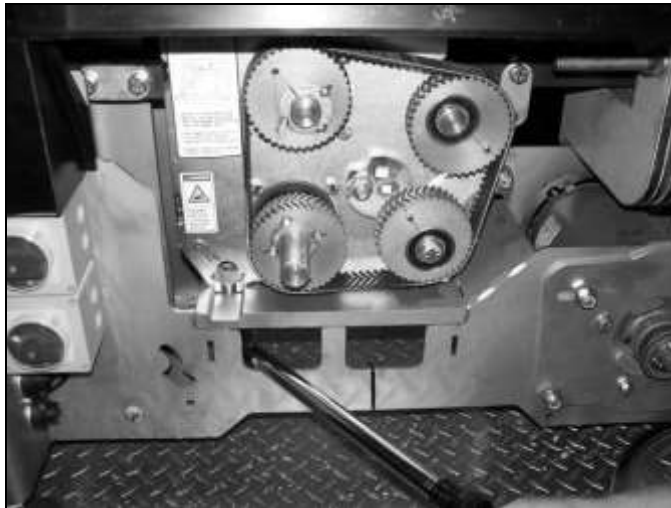


Photo 5-40 – Reinstall four (4) underside bolts

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 13 Loosen the pan connecting rod bushing. Take two full turns off of each bushing clamp bolt using a 14 mm [9/16-inch] socket wrench (see Photo 5-41).



Photo 5-41 – Loosen pan connecting rod bushing

- 14 Loosen the counterweight connecting rod bushing. Using a screwdriver, remove the two (2) screws holding the enclosure panel on the end of the drive enclosure (see Photo 5-42).

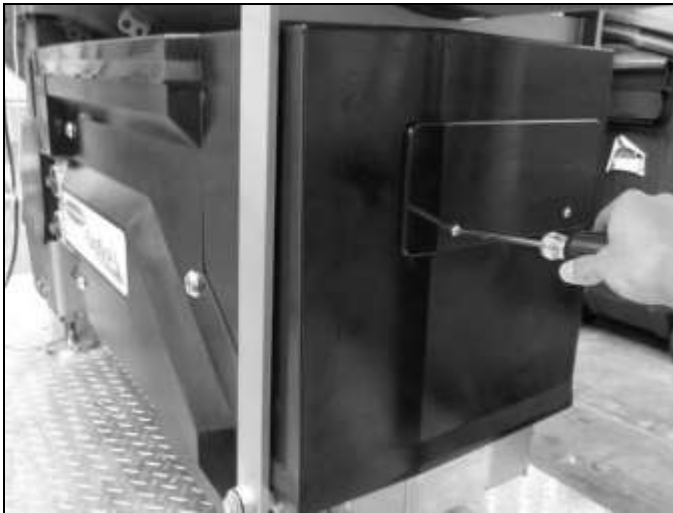


Photo 5-42 – Remove enclosure counterweight access port

Using a 14 mm [9/16-inch] socket wrench, take two full turns off of each counterweight bushing clamp bolt (see Photo 5-43).



Photo 5-43 – Loosen counterweight connecting rod bushing

- 15 Reconnect the pan connecting rods. Using a 19 mm [3/4-inch] socket and torque wrench set at 50 ft-lbs [68 N-m], reinstall the bolts in the pan and counterweight connecting rods located on either side of the transmission (see Photo 5-44).

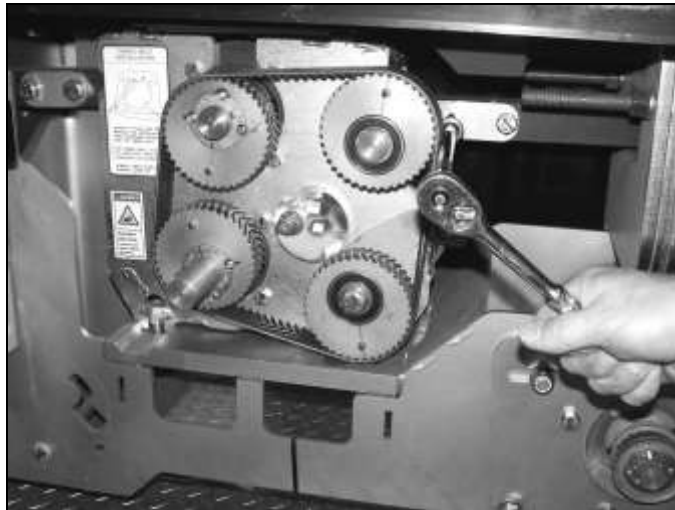


Photo 5-44 – Reconnect the pan and counterweight connecting rods

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 16 Reinstall the drive sprocket and bushing. Using the measurement from step 4, reinstall the drive sprocket, and make sure the key is present on the drive shaft (see Photo 5-45).

Remove the three (3) bolts from the tapped holes in the bushing flange. Reinsert the bolts into the through holes in the bushing flange and align them to the tapped holes in the sprocket.

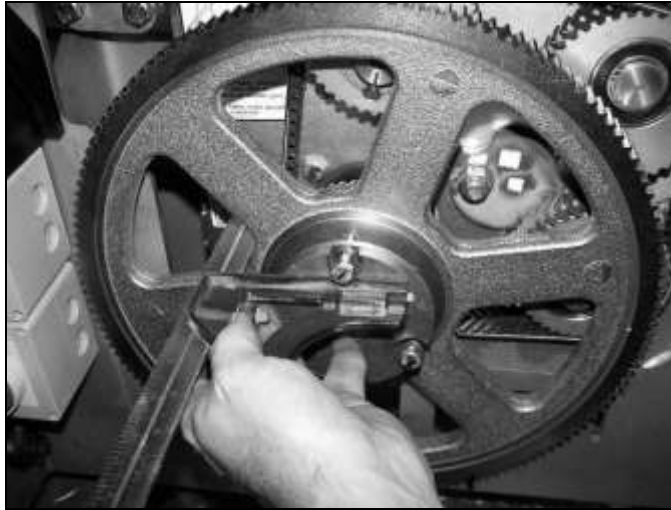


Photo 5-45 – Reinstall drive sprocket bushing using earlier measurement

Using a 1/2-inch [13 mm] socket and torque wrench set to 15 ft-lbs [20 N-m], tighten all three (3) bolts incrementally around the bolt pattern on the drive sprocket bushing (see Photo 5-46).



Photo 5-46 – Tighten the drive sprocket bushing bolts

- 17 Re-clamp the connecting rod bushings to the pan drive arm. Align the “PAN” line of the crankshaft sprocket so that it is horizontal and the word “PAN” is right-side up (see Photo 5-47).

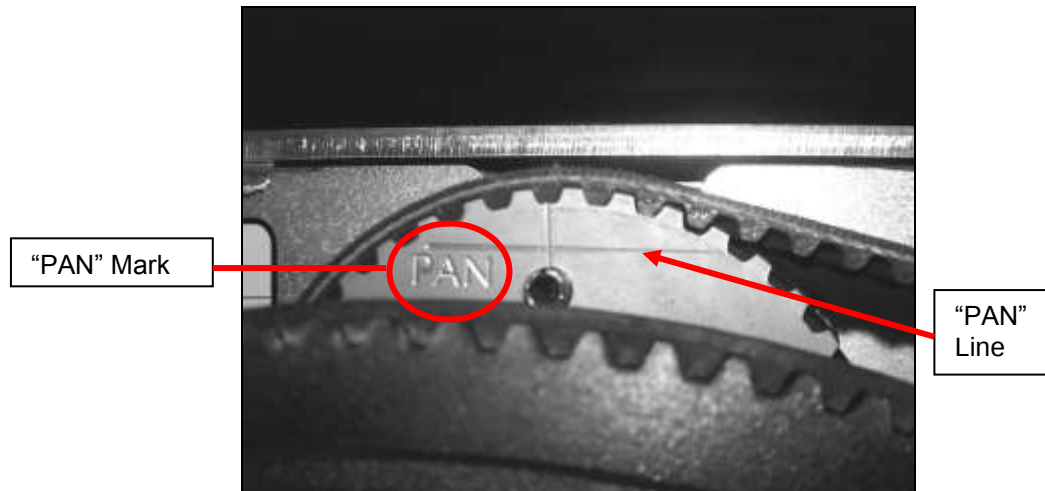


Photo 5-47 – “PAN” line rotated to horizontal position

Using a 14 mm [9/16-inch] socket wrench, evenly tighten the four (4) pan bushing clamp bolts while maintaining an equal gap space between the drive arm and the top and bottom of the clamp (see Photo 5-48).

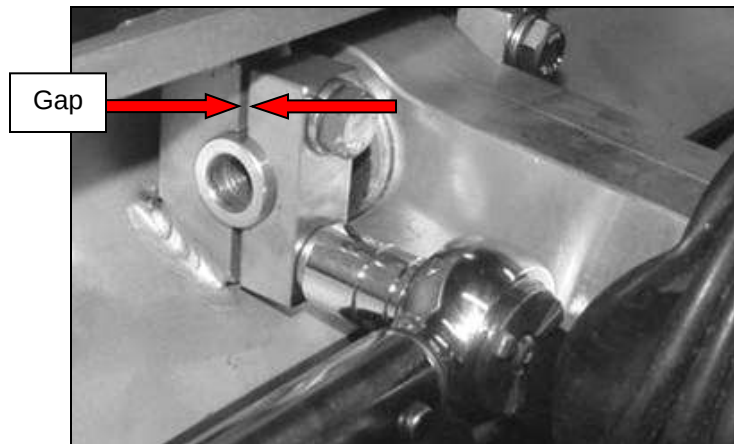


Photo 5-48 – Tighten the bushing clamp bolts on the “PAN” drive arm

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 18 Re-clamp the connecting rod bushings to the counterweight drive arm.
Align the “CW” line of the crankshaft sprocket so that it is horizontal and the word “CW” is right-side up (see Photo 5-49).

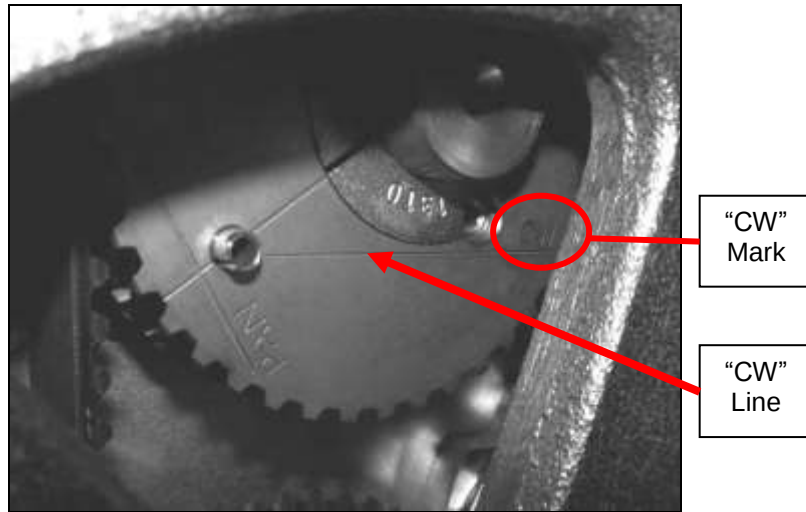


Photo 5-49 – “CW” line rotated to horizontal position

Using a 14 mm [9/16-inch] socket and a torque wrench set at 20 ft-lbs [27 N-m], tighten the four (4) bushing clamp bolts (see Photo 5-50).

Note: *Evenly tighten the four (4) bushing clamp bolts while maintaining an equal gap between the drive arm and the top and bottom of the clamp.*



Photo 5-50 – Tighten the bushing clamp bolts on the “CW” drive arm

Using a #2 Phillips head screwdriver, reinstall the two (2) screws holding the enclosure counterweight access port on the end of the drive enclosure.

- 19 Reinstall the motor belt and tighten as described in “Motor Belt Replacement and Tensioning”.
- 20 Install the enclosure side panel as described in “Removing and Installing Enclosure Side Panel”.

Eccentric Belt Replacement and Tensioning

This procedure describes how to replace and apply the proper tension to the timing belt of the eccentric sprocket system. The eccentric timing belt should receive full tension only if that transmission is to be placed directly into a 260E-G2 drive unit. While in storage, spare transmissions should only maintain enough tension on the belt to maintain the timing. Spare transmissions should have the direction of travel verified, and be fully tensioned before being placed into a 260E-G2 drive unit.

Tools Needed:

- 3/8-inch [10 mm] standard screwdriver
- socket wrench
- torque wrench set at 42 ft-lbs [57 N-m], 50 ft-lbs [68 N-m], & 100 ft-lbs [136 N-m]
- 10-inch [254 mm] long 1/2-inch [12.7 mm] drive socket extension
- 6-inch [152.4 mm] long 3/8-inch [10 mm] drive extension
- 3/4-inch [19 mm] deep well socket
- 3/4-inch [19 mm] socket
- FastBack® 260E-G2 timing plate (part no. 30593041)
- 1-1/16-inch [27 mm] socket
- breaker bar

Step	Description
------	-------------

- | | |
|---|--|
| 1 | Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position. |
|---|--|



WARNING

To prevent the FastBack® from accidentally starting during this procedure, lockout / tag out before servicing.

Moving parts are located behind the enclosure side panel. Do not operate with cover or enclosure side panel removed. Exercise caution if turning the mechanisms by hand to avoid crushing or pinching injuries.

- | | |
|---|---|
| 2 | Remove enclosure side panel as described in "Removing and Installing Enclosure Side Panel". |
| 3 | Remove motor belt as described in "Motor Belt Replacement and Tensioning". |
| 4 | Remove transmission as described in "Transmission Replacement". |

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 5 Loosen the tensioner sprocket pivot nut. Using a 19 mm [3/4-inch] socket, loosen the tensioner sprocket pivot nut one-quarter turn (see Photo 5-51).

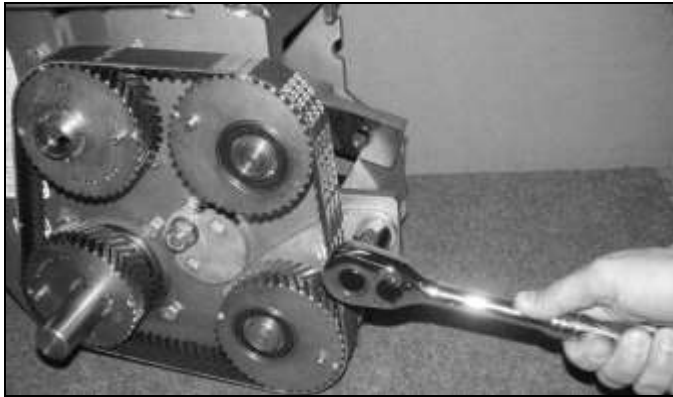


Photo 5-51 – Loosen tensioner sprocket pivot nut

- 6 Loosen the nut behind the tensioner sprocket stud. Using a 3/4-inch [19 mm] socket and breaker bar, loosen the nut behind the tensioner sprocket stud no more than one-half turn (see Photo 5-52).



Photo 5-52 – Loosen nut behind tensioner sprocket stud

- 7 Loosen the nut on the tensioner cam pivot bolt. Using a 19 mm [3/4-inch] socket, loosen the nut on the tensioner cam pivot bolt one-quarter turn (see Photo 5-53).



Photo 5-53 – Loosen tensioner cam pivot nut

Note: *If only re-tensioning the belt, skip to step 11 of this procedure.*

- 8 Remove the eccentric timing belt. Push up on the tensioner sprocket by hand to create enough slack in the eccentric belt to remove it (see Photo 5-54). Inspect eccentric timing belt for cracks and excessive wear. See Appendix B for the belt's part number.

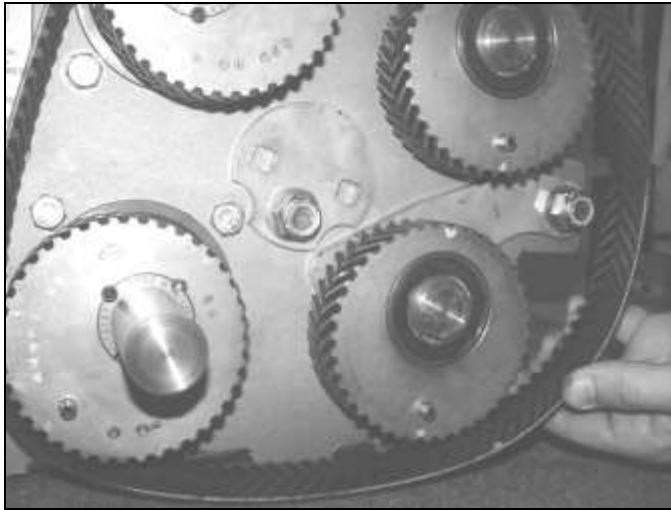


Photo 5-54 – Remove eccentric timing belt

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 9 Time the crankshaft and drive shaft sprockets by placing the timing plate (part no. 30593041) over the ends of their shafts and aligning the pins in the slots of the plate as shown in Photo 5-55.

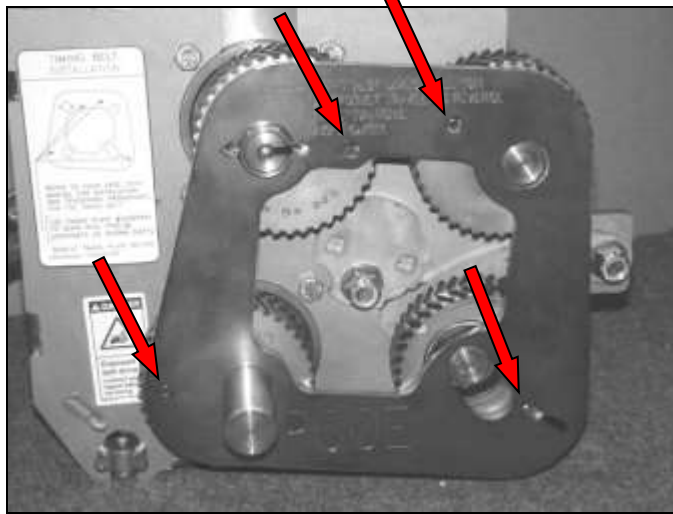


Photo 5-55 – Correct placement of timing plate

Note: Check the direction of product flow. This is indicated when the slot on the end of the crankshaft lines up with either of the arrow-shaped cutouts. The transmission shown in Photo 5-56 will convey product to the right. If the direction of product travel is not correct for the conveyor receiving the transmission, refer to “Changing the Direction of Product Flow”.

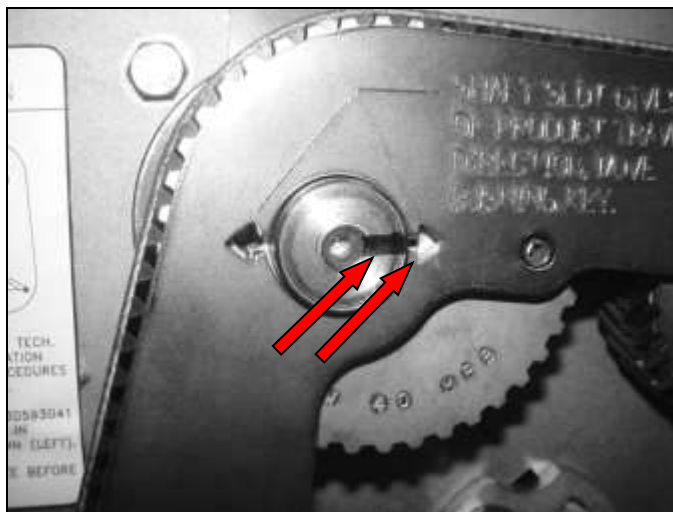


Photo 5-56 – Check that product flow is in the correct direction

Orient the tensioner and idler sprockets in the same way (see Photo 5-56). Ensure that the tensioner sprocket is rotated upward in its slot and that the nut on the tensioner sprocket stud is no more than one-half turn loose.

- 10 Slide the eccentric timing belt over all four sprockets. Make sure there is no slack between the crankshaft, drive, and idler sprockets (see Photo 5-57).



Photo 5-57 – Make sure there is no slack in the belt

- 11 Remove slack from the eccentric belt. Push down on the tensioner sprocket by hand, and apply a clockwise torque to the tensioner cam until all the slack is removed from the eccentric belt.
- The pin in the tensioner sprocket will be in the lower end of its slot in the timing plate, and the belt tension on either side of the tensioner sprocket should be the same.
- Using the 19 mm [3/4-inch] socket, tighten the tensioner pivot nut slightly to hold the tensioner in place and remove the timing plate (see Photo 5-58).



Photo 5-58 – Remove slack from eccentric belt

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 12 Tighten the eccentric sprocket system. Using the 19 mm [3/4-inch] socket with a torque wrench set at 50 ft-lbs [68 N-m] and the 6-inch [152.4 mm] long 3/8-inch [10 mm] drive extension, torque the tensioner cam until the torque wrench reaches its set point (see Photo 5-59).



Photo 5-59 – Tighten the tensioner cam pivot nut

While holding the cam with 50 ft-lbs [68 N-m], tighten the tensioner cam pivot nut using a 19 mm [3/4-inch] socket (see Photo 5-60).

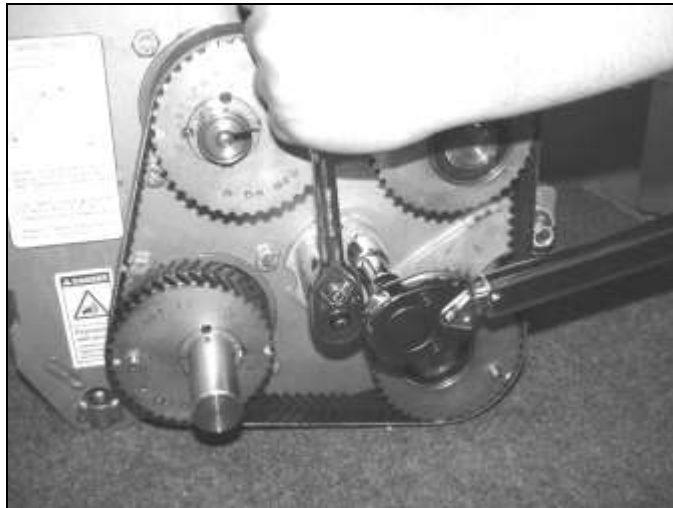


Photo 5-60 – Torque the tensioner cam pivot nut

Using the 19 mm [3/4-inch] socket and the torque wrench set at 42 ft-lbs [57 N-m], tighten the tensioner pivot nut (see Photo 5-61).

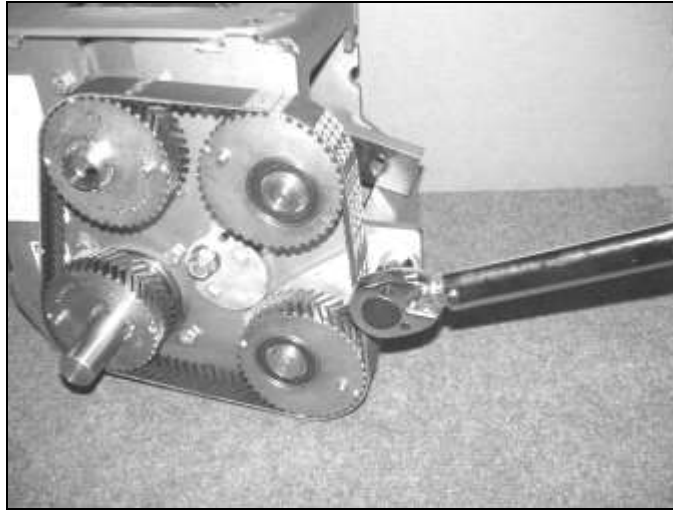


Photo 5-61 – Torque the tensioner sprocket pivot nut

Using the 3/4-inch [19 mm] socket and a torque wrench set at 100 ft-lbs [136 N-m], tighten the nut on the back of the tensioner sprocket stud (see Photo 5-62).



Photo 5-62 – Torque the nut on the tensioner sprocket stud

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 13 Follow the three steps illustrated in Figure 5-1 to ensure proper belt timing:
 1. Starting at timing pin, count twenty (20) sprocket grooves clockwise.
 2. From there, count eighteen (18) teeth on belt.
 3. The eighteenth (18th) tooth should align with the machined line in the idler sprocket.

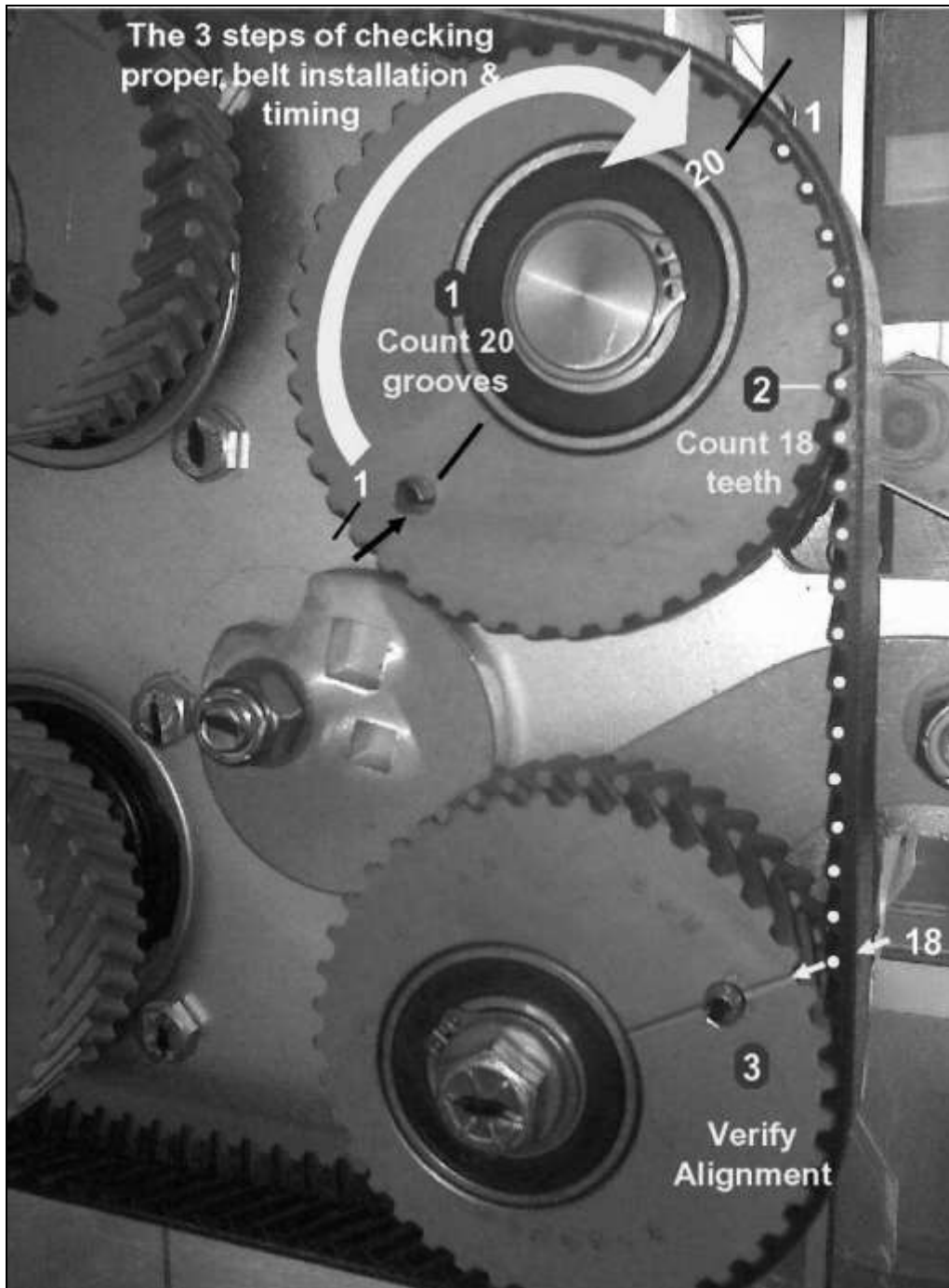


Figure 5-1 – Three steps to ensure proper belt timing

- 14 Reinstall the transmission as described in “Transmission Replacement”, continuing from step number 10, “Install new transmission”.
Note: *If returning the transmission to the original drive, then skip steps 13, 14, and 15, (Loosen clamp connecting rod bushings) and steps 17 and 18 (Tighten clamp connecting rod bushings).*
- 15 Remove the transmission removal plate.
- 16 Tighten the motor belt as described in “Motor Belt Replacement and Tensioning”.
- 17 Install the enclosure side panel as described in “Removing and Installing Enclosure Side Panel”.

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

Checking/Reversing the Direction of Product Flow

This procedure describes how to check and, if desired, reverse the direction of product flow via the transmission. This procedure should be performed before a spare transmission is placed in a 260E-G2 drive unit.

Note: *Reversing the motor will not reverse the direction of product flow.*

Tools Needed:

- 3/8-inch [10 mm] standard screwdriver
- torque wrench set at 55 in-lbs [6.2 N-m]
- straight edge
- FastBack® 260E-G2 timing plate (part no. 30593041)
- 3/16-inch [4.5 mm] allen wrench

Step	Description
------	-------------

- | | |
|---|--|
| 1 | Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position. |
|---|--|



WARNING

To prevent the FastBack® from accidentally starting during this procedure, lockout / tag out before servicing.

Moving parts are located behind the enclosure side panel. Do not operate with cover or enclosure side panel removed. Exercise caution if turning the mechanisms by hand to avoid crushing or pinching injuries.

- | | |
|---|--|
| 2 | Remove the enclosure side panel as described in “Removing and Installing Enclosure Side Panel”. |
| 3 | Rotate the main drive sprocket so that the crankshaft sprocket timing pin is pointing towards the center line of the transmission. |

The direction of product flow is then noted by the slot on the end of the crankshaft.

Case #1: If the slot is closer to the timing pin, then the direction of product flow will be towards the counterweight assembly (see Photo 5-63).

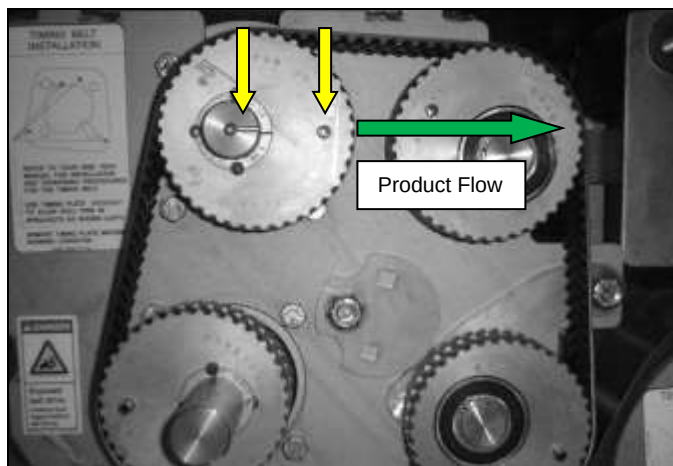


Photo 5-63 – Example of Case #1

Case #2: If the slot is at its farthest point from the timing pin, then the direction of the product flow will be towards the drive arm assembly (see Photo 5-64).

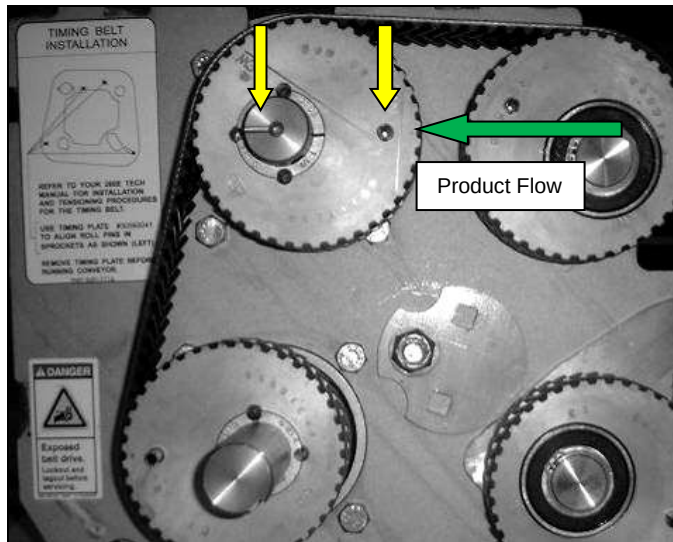


Photo 5-64 – Example of Case #2

Note: *If product flow is in the desired direction, then stop here. If product flow is desired in the opposite direction, then continue with this maintenance procedure.*

- 4 Remove the transmission as described in "Transmission Replacement" and place it on a clean, well-supported working surface.
- 5 Remove the eccentric belt as described in "Eccentric Belt Replacement and Tensioning".
- 6 Using a permanent marker, note the location of the bushing along the crankshaft. This allows for easier replacement of the sprocket later in this procedure. Using a 3/16-inch [4.5 mm] allen wrench, remove crankshaft sprocket set screws (see Photo 5-65).

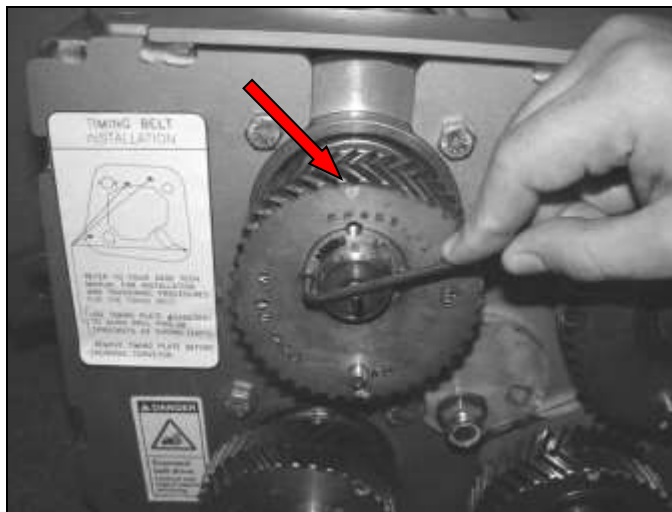


Photo 5-65 – Bushing location marked on crankshaft sprocket

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

Insert one (1) of the set screws into the middle hole and slowly tighten the screw until the taper lock bushing is removed from the crankshaft sprocket (see Photo 5-66). Carefully note in which crankshaft key slot the bushing key was originally housed.

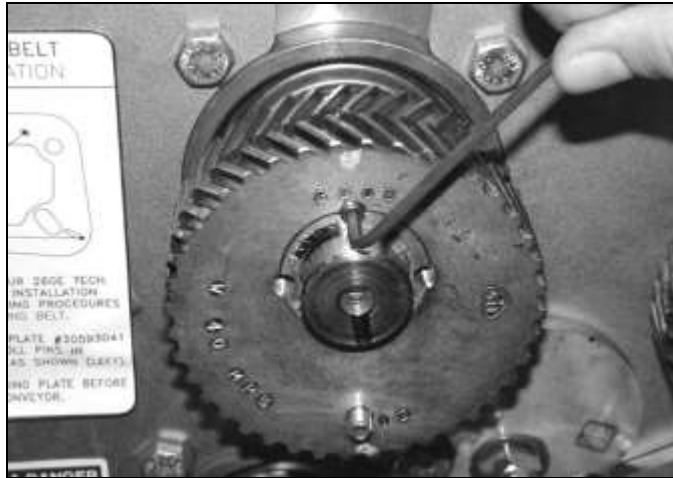


Photo 5-66 – Insert set screw in middle hole to remove sprocket

- 7 Reverse the direction of product flow. Remove the bushing key from original slot in crankshaft, and insert it into slot on the opposite side of crankshaft.
- 8 Install and align crankshaft sprocket on the crankshaft. Using a 3/16-inch [4.5 mm] allen wrench and a torque wrench, insert the set screws and tighten in equal increments to 55 in-lbs [6.2 N-m] (see Photo 5-67).



Photo 5-67 – Torque the set screws of the crankshaft sprocket

Ensure that the crankshaft sprocket is in proper alignment with the two adjacent transmission sprockets using a straight edge (see Photo 5-68).

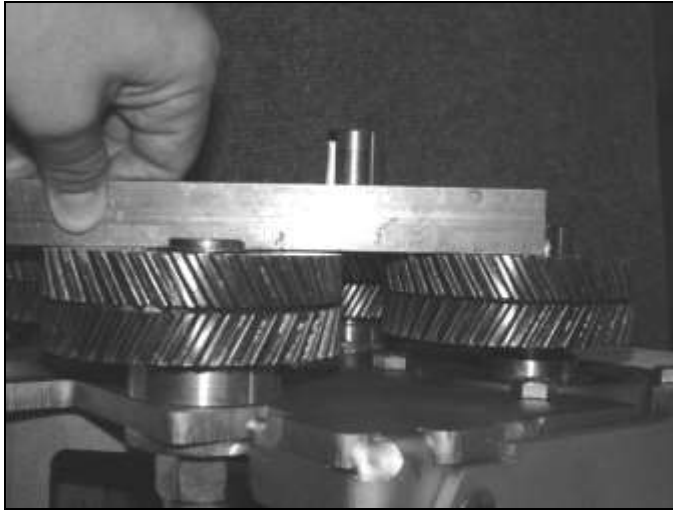


Photo 5-68 – Check the alignment of the sprockets

Verify the product flow direction using the method described in step 3 or by placing the timing plate over the crankshaft sprocket and lining up the pin in the slot (timing plate part no. 30593041) (see Photo 5-69).

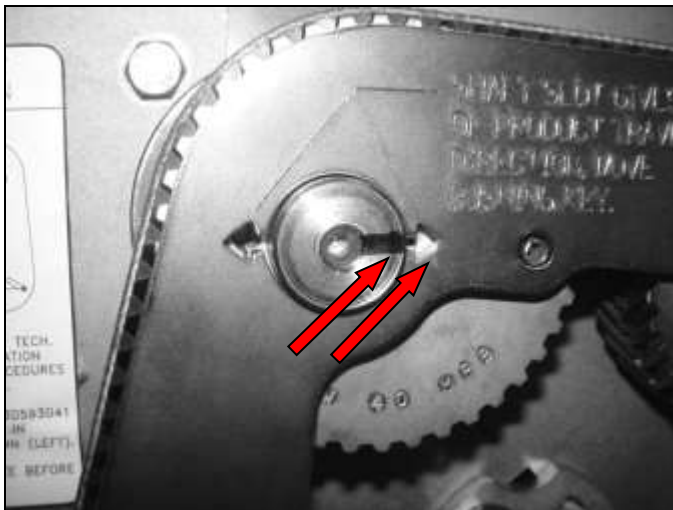


Photo 5-69 – Verify that product flow is in the desired direction

- 9 Reinstall the eccentric belt as described in "Eccentric Belt Replacement and Tensioning".
- 10 Reinstall the transmission as described in "Transmission Replacement".
- 11 Tighten the motor belt as described in "Motor Belt Replacement and Tensioning".
- 12 Install the enclosure side panel as described in "Removing and Installing Enclosure Side Panel".

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

Adding / Removing Counterweight Plates

This procedure describes how to add or remove counterweight plates from the base counterweight billet. This procedure should be performed when there is a significant increase or decrease to the pan's weight.

Only a certified Heat and Control engineering modification drawing should be utilized for a counterweight addition/subtraction. Please contact a Heat and Control representative for this custom-designed modification.

Tools Needed:

- 3/8-inch [10 mm] standard screwdriver
- socket wrench
- 14 mm [9/16-inch] socket
- torque wrench set at 20 ft-lbs [27 N-m]
- 10-inch [254 mm] long 1/2-inch [12.7 mm] drive socket extension
- 19 mm [3/4-inch] socket
- 15/16-inch [24 mm] open-ended wrench

Step	Description
------	-------------

- | | |
|---|--|
| 1 | Disconnect power to the FastBack® by rotating the emergency disconnect switch to the OFF position. |
|---|--|



WARNING

To prevent the FastBack® from accidentally starting during this procedure, lockout / tag out before servicing.

Moving parts are located behind the enclosure side panel. Do not operate with cover or enclosure side panel removed. Exercise caution if turning the mechanisms by hand to avoid crushing or pinching injuries.

- | | |
|---|---|
| 2 | Remove the enclosure side panel as described in "Removing and Installing Enclosure Side Panel". |
| 3 | Remove the motor belt as described in "Motor Belt Replacement and Tensioning". |

- 4 Remove the counterweight connecting rod bolts located to the right of the transmission using a 19 mm [3/4-inch] socket wrench with extension (see Photo 5-70).

Note: *Rotate the eccentric sprocket system by hand to achieve the greatest access to the bolts.*



Photo 5-70 – Remove counterweight connecting rod bolts at transmission

- 5 Remove the counterweight connecting rod bolts at the counterweight drive arm. Using a 14 mm [9/16-inch] socket wrench, remove each counterweight bushing clamp bolt (see Photo 5-71).



Photo 5-71 – Remove counterweight connecting rod bolts at counterweights

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 6 Remove the counterweight connecting rod. Pull counterweight connecting rod out through the enclosure counterweight access port (see Photo 5-72).



Photo 5-72 – Remove counterweight connecting rod

- 7 Remove the counterweight plate nuts. Using a 15/16-inch [24 mm] open-ended wrench, remove the two (2) counterweight plate nuts that are pressing the stack of counterweight plates together on the counterweight bolts (see Photo 5-73). Secure other end of bolt from the enclosure counterweight access port as needed.

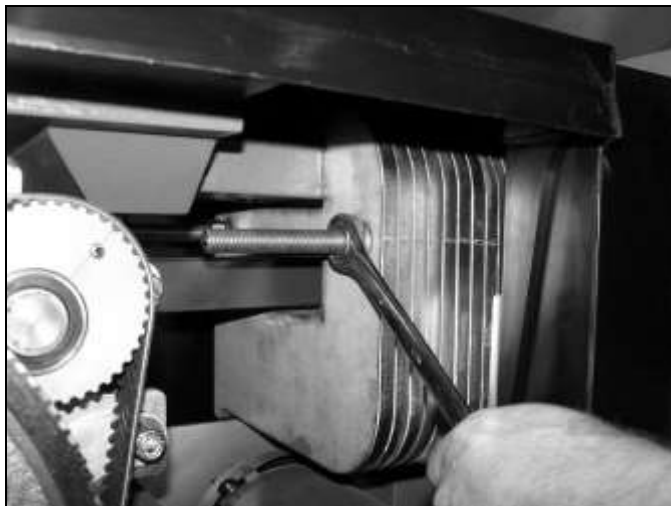


Photo 5-73 – Remove counterweight plate nuts

- 8 Add or remove counterweight plates, as specified by a Heat and Control modification drawing. Either add or remove counterweight plates by sliding them onto the counterweight casting (see Photo 5-74).

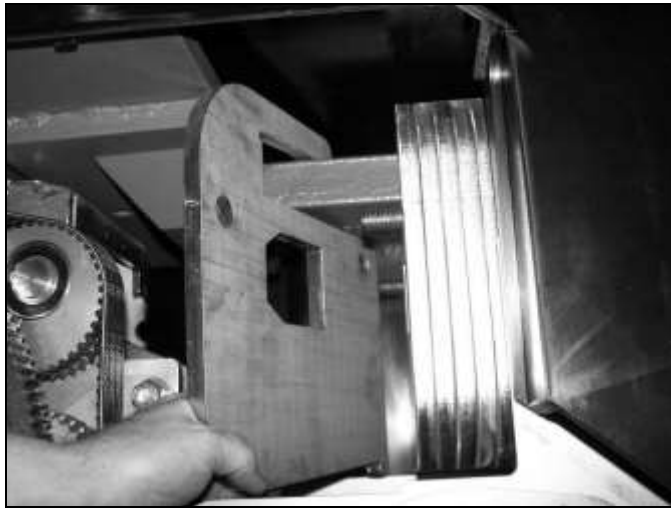


Photo 5-74 – Add/remove counterweight plates

- 9 Reinstall the counterweight plate nuts. Using a 15/16-inch [23/24 mm] open-ended wrench, reinstall the two (2) counterweight plate nuts (see Photo 5-75). Secure other end of bolt from the enclosure counterweight access port as needed.



Photo 5-75 – Reinstalling counterweight plate nuts

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

- 10 Reinstall the counterweight connecting rod. Push counterweight connecting rod in through the enclosure counterweight access port and then through the stack of counterweight plates (see Photo 5-76).



Photo 5-76 – Reinstall counterweight connecting rod

- 11 Reinstall the counterweight connecting rod bolts at the transmission. Using a 19 mm [3/4-inch] socket wrench with extension, reinstall the counterweight connecting rod bolts located to the right of the transmission (see Photo 5-77).

Note: *Rotate the eccentric sprocket system by hand to achieve the greatest access to the bolts.*

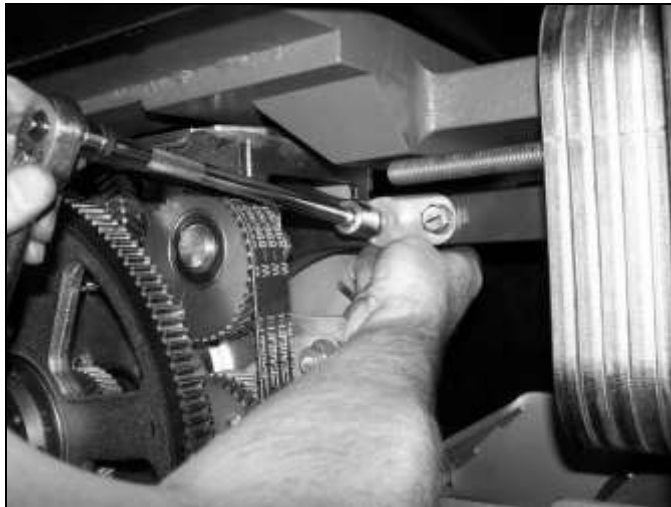


Photo 5-77 – Reinstall counterweight connecting rod bolts at transmission

- 12 Reinstall the counterweight connecting rod bolts at the counterweight drive arm. Using a 14 mm [9/16-inch] socket and a torque wrench set at 20 ft-lbs [27 N-m], tighten the four (4) counterweight connecting rod bolts (see Photo 5-78).

Note: *Evenly tighten the four (4) counterweight connecting rod bolts while maintaining an equal gap between the drive arm and the top and bottom of the clamp.*



Photo 5-78 – Tighten the counterweight connecting rod bolts at counterweights

- 13 Reinstall the motor belt and tighten as described in “Motor Belt Replacement and Tensioning”.
- 14 Install the enclosure side panel as described in “Removing and Installing Enclosure Side Panel”.

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

VFD Replacement

This procedure describes the necessary steps to change the VFD(s).

Step	Description
------	-------------

- | | |
|---|---|
| 1 | <p>Disconnect power to the FastBack® by rotating the emergency disconnect switch(es) to the OFF position.</p> <p>Disable power provided to the FastBack® system at the source by turning off the power to the power distribution panel.</p> |
|---|---|



WARNING

Allow system a cool-down time of 30 seconds to avoid any chance of electrocution.

To prevent the FastBack® from accidentally starting during this procedure, lockout / tag out before servicing.

Moving parts are located behind the enclosure side panel. Do not operate with cover or enclosure side panel removed. Exercise caution if turning the mechanisms by hand to avoid crushing or pinching injuries.

- | | |
|---|---|
| 2 | <p>Using a #2 Phillips head screwdriver, remove the utility enclosure lid (see Photo 5-79).</p> |
|---|---|



Photo 5-79 – Remove power module lid

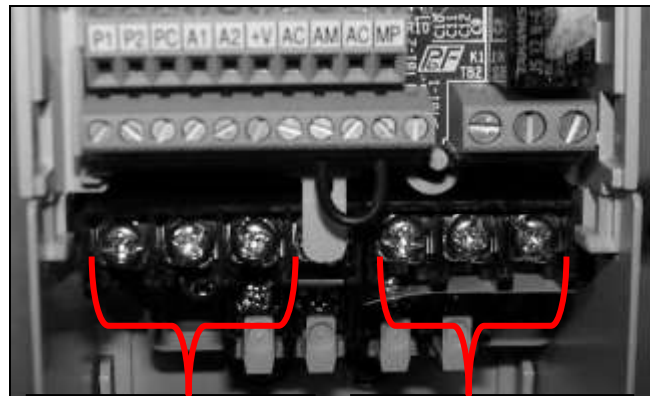
- | | |
|---|---|
| 3 | <p>Disconnect the communication jumper from the inverter by pressing the tab on the jack and pulling out.</p> |
|---|---|

- 4 Use a #2 Phillips head screwdriver to remove the front cover of inverter by loosening the screw and pulling the cover off (see Photo 5-80).



Photo 5-80 – Control panel cover removed

- 5 Using a #2 Phillips head screwdriver, disconnect the motor leads running to the T1, T2, and T3 terminals of the inverter (see Photo 5-81).
Remove the main power leads running to L1, L2, and L3 terminals of the inverter (see Photo 5-81).



Leads L1, L2, and
L3 to inverter

Leads T1, T2, and
T3 to motor

Photo 5-81 – Power Leads

K-5 – MAINTENANCE

FastBack® Model 260E-G2 Technical Manual

Remove the control wires from the green terminal blocks (see Photo 5-82). Mark or label the wires as necessary.

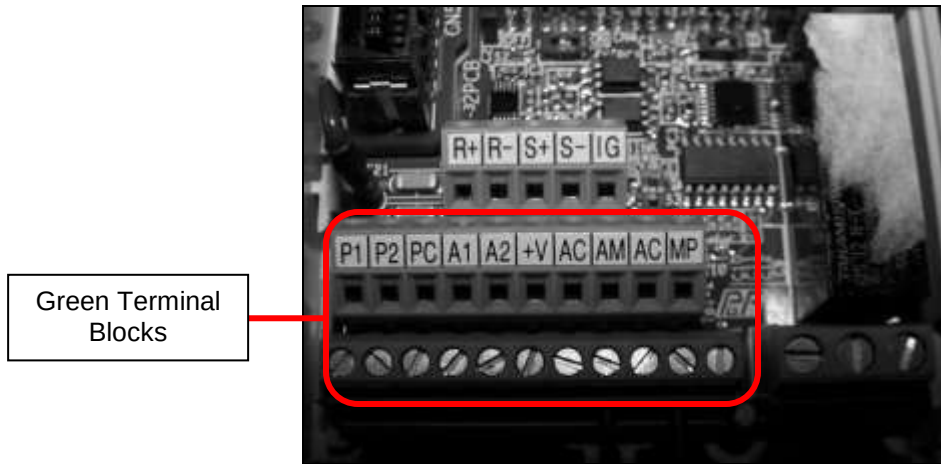


Photo 5-82 – Control Wires

- 6 Using a 3 mm allen wrench, remove the four (4) fasteners that secure the VFD to the back panel.
Remove the VFD and ship back to Heat and Control (if still under warranty).
Mount new VFD in place and secure with fasteners.
- 7 Using a #2 Phillips head screwdriver, connect the motor leads to T1, T2, and T3 terminals of the inverter.
Using a #2 Phillips head screwdriver, connect the main power leads to L1, L2, and L3 terminals of the inverter.
Using an 11/32-inch [8.73 mm] socket wrench, attach the grounding straps to the wall of the enclosure.
Attach the control wires to the appropriate green terminal block locations as noted in step number 5 of this procedure (see Photo 5-82).
- 8 Restore power to the FastBack® system at the source by returning power to the power distribution panel.
Restore power to the FastBack® by rotating the disconnect switch to the ON position.
- 9 Initialize the new VFD using the procedure found in **Technical Publication PH009** (Yaskawa) or **Technical Publication PH015** (Allen-Bradley).
- 10 Using the FastBack® hand-held operator, initiate a run sequence and ensure that the motor turns in a counterclockwise direction.
If the motor does not run in the proper direction, then turn off the power to the FastBack® system and swap any two leads between “T1”, “T2”, and “T3”.

- 11 Install the inverter front cover. Use a #2 Phillips head screwdriver to reinstall the front cover of inverter (see Photo 5-83).

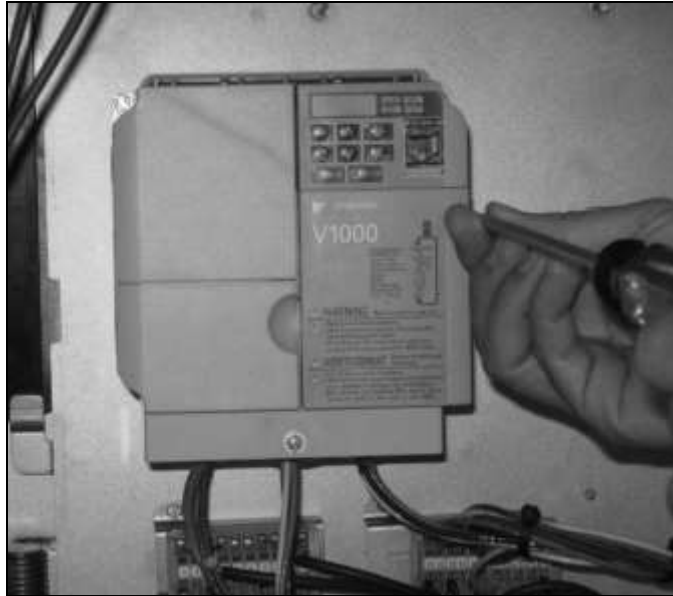


Photo 5-83 – Reinstall control panel cover

- 12 Ensure the communication jumper is inserted into the connection correctly, with the leads of the cable facing the leads of the connector.
- 13 Using a 1/4-inch [6.35 mm] standard screwdriver, reinstall the utility enclosure lid.